

dml

Shun

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Run DML Analysis

```
##
## === Health & Social | n=1987 person-waves, 1508 persons ===
##   volunteered  x LASSO      ...

## ATE= 0.0533  SE=0.0255
##   volunteered  x SVM        ...

## ATE= 0.0521  SE=0.0279
##   health_score x LASSO      ... ATE= 0.0037  SE=0.0114
##   health_score x SVM        ...

## ATE= 0.0112  SE=0.0119
##   social_media x LASSO      ... ATE=-0.0169  SE=0.0117
##   social_media x SVM        ...

## ATE=-0.0152  SE=0.0119
##
## === Responsibility Attitudes | n=1973 person-waves, 1498 persons ===
##   resp_earn    x LASSO      ... ATE= 0.1073  SE=0.0238
##   resp_earn    x SVM        ...

## ATE= 0.0902  SE=0.0249
##   resp_rent    x LASSO      ... ATE= 0.0159  SE=0.0209
##   resp_rent    x SVM        ...

## ATE= 0.0362  SE=0.0219
##   resp_bills   x LASSO      ... ATE= 0.0507  SE=0.0249
##   resp_bills   x SVM        ...

## ATE= 0.0553  SE=0.0259
##   resp_money   x LASSO      ... ATE=-0.0990  SE=0.0215
##   resp_money   x SVM        ...

## ATE=-0.0203  SE=0.0269
##
## === Lagged Wellbeing | n=1973 person-waves, 1498 persons ===
##   wb_emo_lag    x LASSO      ... ATE=-0.0075  SE=0.0134
##   wb_emo_lag    x SVM        ...
```

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## ATE=-0.0002 SE=0.0139
##   wb_soc_lag   x LASSO   ... ATE= 0.0005 SE=0.0107
##   wb_soc_lag   x SVM     ...

## ATE= 0.0062 SE=0.0110
##   wb_psy_lag   x LASSO   ... ATE= 0.0116 SE=0.0130
##   wb_psy_lag   x SVM     ...

## ATE= 0.0207 SE=0.0134
##   sx_anx_lag   x LASSO   ... ATE=-0.0190 SE=0.0114
##   sx_anx_lag   x SVM     ...

## ATE=-0.0133 SE=0.0118
##
##
## ===== NUISANCE MSE COMPARISON (lower = better) =====
##           group      treatment learner mse_outcome mse_treatment
##           Health & Social health_score LASSO      0.2165      0.9421
##           Health & Social health_score SVM         0.2334      0.9393
##           Health & Social social_media LASSO      0.2171      0.9282
##           Health & Social social_media SVM         0.2330      0.9883
##           Health & Social volunteered LASSO      0.2201      0.5022
##           Health & Social volunteered SVM         0.2330      0.5072
##           Lagged Wellbeing sx_anx_lag LASSO      0.2150      0.8854
##           Lagged Wellbeing sx_anx_lag SVM         0.2295      0.8959
##           Lagged Wellbeing wb_emo_lag LASSO      0.2150      0.6678
##           Lagged Wellbeing wb_emo_lag SVM         0.2302      0.6831
##           Lagged Wellbeing wb_psy_lag LASSO      0.2149      0.6687
##           Lagged Wellbeing wb_psy_lag SVM         0.2301      0.6791
##           Lagged Wellbeing wb_soc_lag LASSO      0.2150      0.9745
##           Lagged Wellbeing wb_soc_lag SVM         0.2299      0.9847
##           Responsibility Attitudes resp_bills LASSO 0.2149      0.1860
##           Responsibility Attitudes resp_bills SVM   0.2314      0.1974
##           Responsibility Attitudes resp_earn LASSO 0.2152      0.1954
##           Responsibility Attitudes resp_earn SVM   0.2320      0.2172
##           Responsibility Attitudes resp_money LASSO 0.2150      0.2160
##           Responsibility Attitudes resp_money SVM   0.2316      0.1659
##           Responsibility Attitudes resp_rent LASSO 0.2148      0.2558
##           Responsibility Attitudes resp_rent SVM   0.2306      0.2747
##
## ===== ATE FROM BEST LEARNER PER TREATMENT =====
##           group      treatment learner      ATE      SE p_value sig
##           Health & Social health_score LASSO 0.0037 0.0114 0.7445
##           Health & Social social_media LASSO -0.0169 0.0117 0.1509
##           Health & Social volunteered LASSO 0.0533 0.0255 0.0369 **
##           Lagged Wellbeing sx_anx_lag LASSO -0.0190 0.0114 0.0958 *
##           Lagged Wellbeing wb_emo_lag LASSO -0.0075 0.0134 0.5745
##           Lagged Wellbeing wb_psy_lag LASSO 0.0116 0.0130 0.3744
##           Lagged Wellbeing wb_soc_lag LASSO 0.0005 0.0107 0.9624
##           Responsibility Attitudes resp_bills LASSO 0.0507 0.0249 0.0415 **
##           Responsibility Attitudes resp_earn LASSO 0.1073 0.0238 0.0000 ***
##           Responsibility Attitudes resp_money LASSO -0.0990 0.0215 0.0000 ***
##           Responsibility Attitudes resp_rent LASSO 0.0159 0.0209 0.4460

```

```
##
## Saved: data/processed/tables/dml_all_results.csv
## Saved: data/processed/tables/dml_best_results.csv
```

Main Results

Table 1: Best DML Results by Treatment

	group	treatment	learner	ATE	SE	p_value	sig
Health & Social health_score	Health & Social	health_score	LASSO	0.0037	0.0114	0.7445	
Health & Social social_media	Health & Social	social_media	LASSO	-0.0169	0.0117	0.1509	
Health & Social volunteered	Health & Social	volunteered	LASSO	0.0533	0.0255	0.0369	**
Lagged Wellbeing sx_anx_lag	Lagged Wellbeing	sx_anx_lag	LASSO	-0.0190	0.0114	0.0958	*
Lagged Wellbeing wb_emo_lag	Lagged Wellbeing	wb_emo_lag	LASSO	-0.0075	0.0134	0.5745	
Lagged Wellbeing wb_psy_lag	Lagged Wellbeing	wb_psy_lag	LASSO	0.0116	0.0130	0.3744	
Lagged Wellbeing wb_soc_lag	Lagged Wellbeing	wb_soc_lag	LASSO	0.0005	0.0107	0.9624	
Responsibility Attitudes resp_bills	Responsibility Attitudes	resp_bills	LASSO	0.0507	0.0249	0.0415	**
Responsibility Attitudes resp_earn	Responsibility Attitudes	resp_earn	LASSO	0.1073	0.0238	0.0000	***
Responsibility Attitudes resp_money	Responsibility Attitudes	resp_money	LASSO	-0.0990	0.0215	0.0000	***
Responsibility Attitudes resp_rent	Responsibility Attitudes	resp_rent	LASSO	0.0159	0.0209	0.4460	

Full Results

Table 2: All DML Estimates

group	treatment	learner	ATE	SE	p_value
Health & Social	volunteered	LASSO	0.0533	0.0255	0.0369
Health & Social	volunteered	SVM	0.0521	0.0279	0.0616
Health & Social	health_score	LASSO	0.0037	0.0114	0.7445
Health & Social	health_score	SVM	0.0112	0.0119	0.3465
Health & Social	social_media	LASSO	-0.0169	0.0117	0.1509
Health & Social	social_media	SVM	-0.0152	0.0119	0.2017
Responsibility Attitudes	resp_earn	LASSO	0.1073	0.0238	0.0000
Responsibility Attitudes	resp_earn	SVM	0.0902	0.0249	0.0003
Responsibility Attitudes	resp_rent	LASSO	0.0159	0.0209	0.4460
Responsibility Attitudes	resp_rent	SVM	0.0362	0.0219	0.0980
Responsibility Attitudes	resp_bills	LASSO	0.0507	0.0249	0.0415
Responsibility Attitudes	resp_bills	SVM	0.0553	0.0259	0.0328
Responsibility Attitudes	resp_money	LASSO	-0.0990	0.0215	0.0000

group	treatment	learner	ATE	SE	p_value
Responsibility Attitudes	resp_money	SVM	-0.0203	0.0269	0.4491
Lagged Wellbeing	wb_emo_lag	LASSO	-0.0075	0.0134	0.5745
Lagged Wellbeing	wb_emo_lag	SVM	-0.0002	0.0139	0.9857
Lagged Wellbeing	wb_soc_lag	LASSO	0.0005	0.0107	0.9624
Lagged Wellbeing	wb_soc_lag	SVM	0.0062	0.0110	0.5765
Lagged Wellbeing	wb_psy_lag	LASSO	0.0116	0.0130	0.3744
Lagged Wellbeing	wb_psy_lag	SVM	0.0207	0.0134	0.1220
Lagged Wellbeing	sx_anx_lag	LASSO	-0.0190	0.0114	0.0958
Lagged Wellbeing	sx_anx_lag	SVM	-0.0133	0.0118	0.2589

Figure 1: ATE Estimates

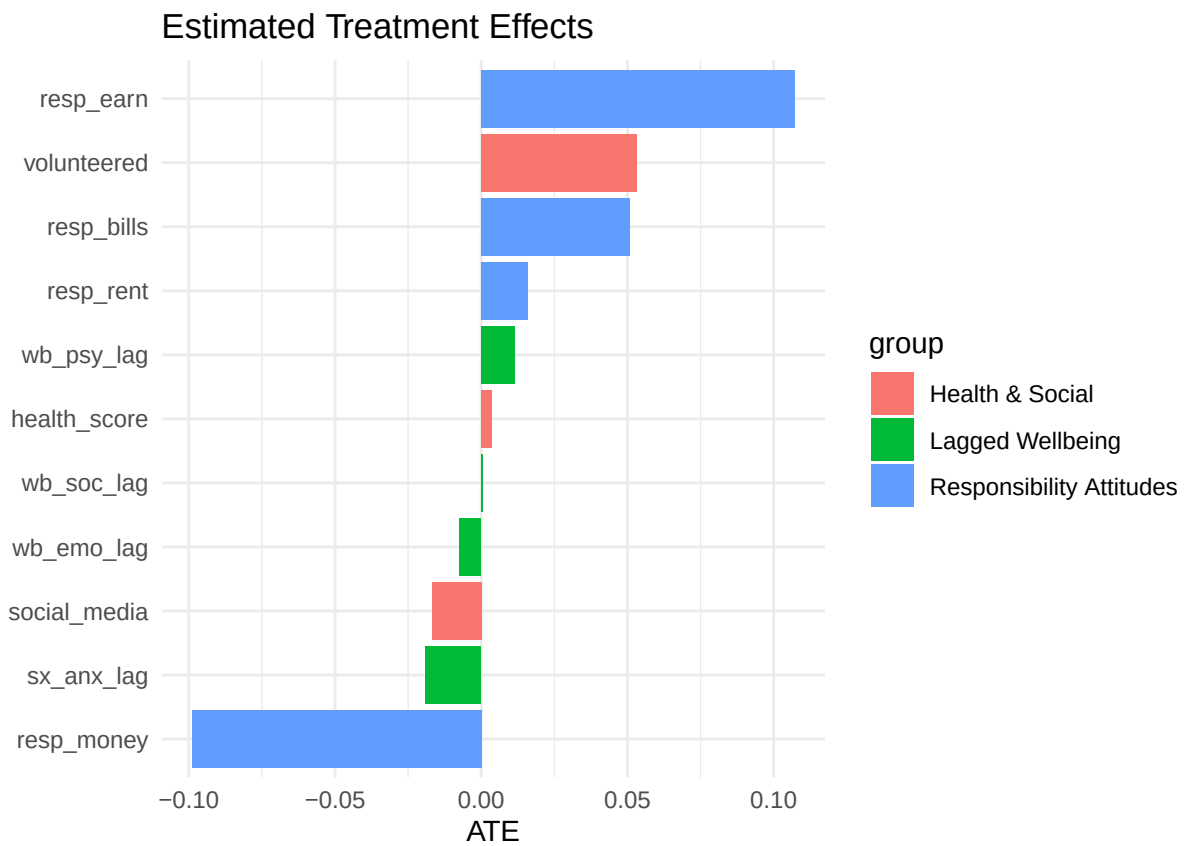


Figure 2: P-values

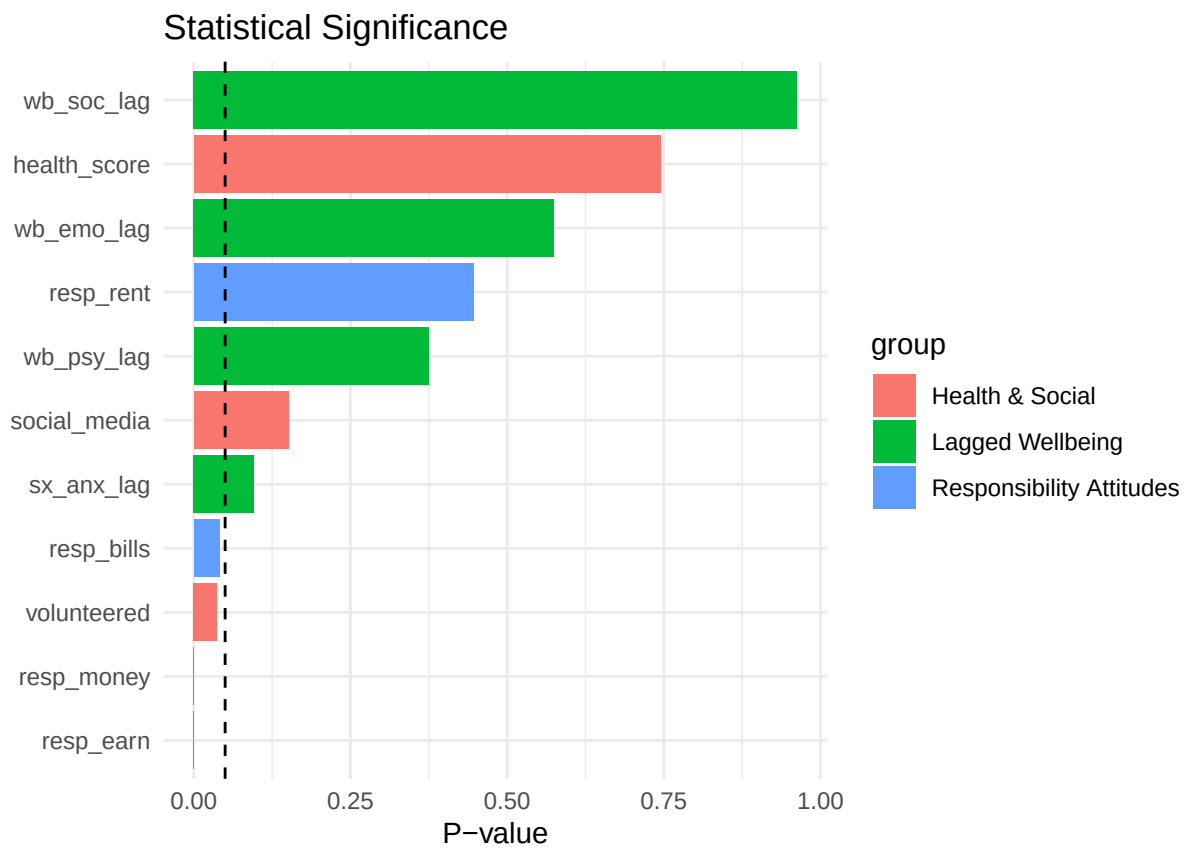
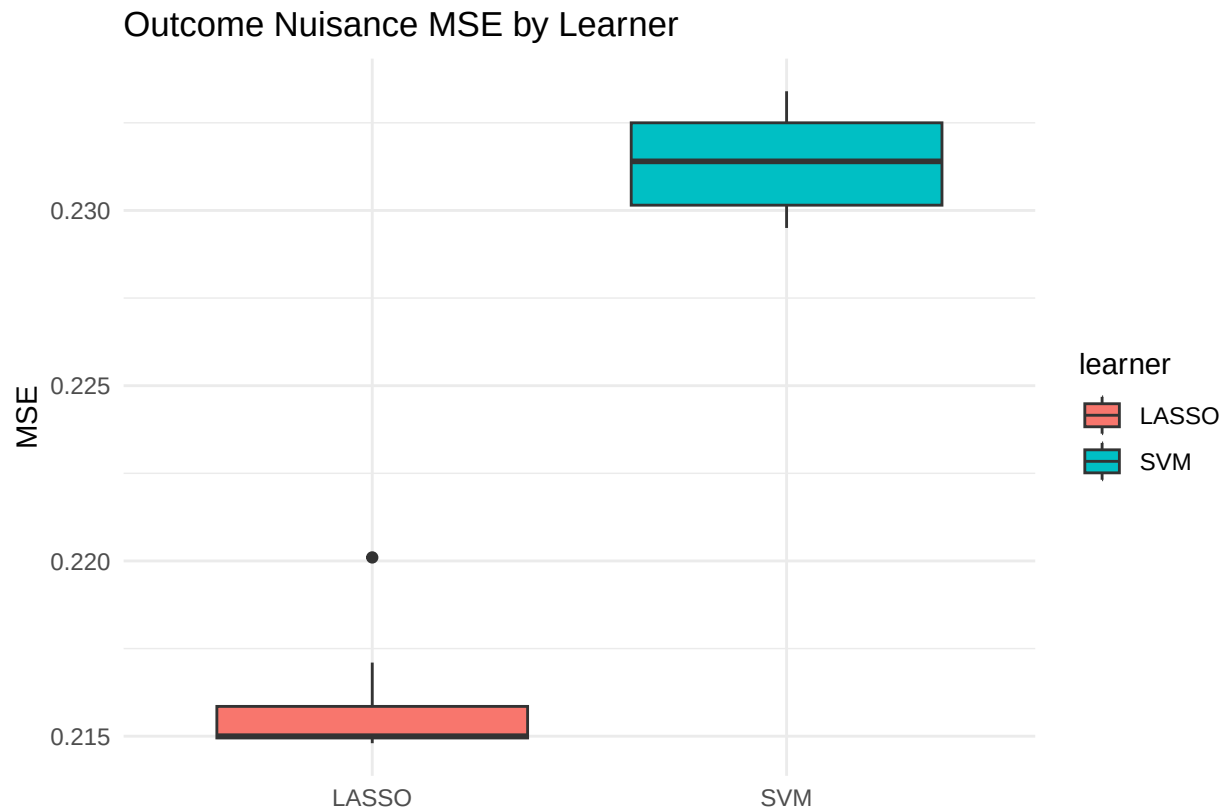


Figure 3: Model Comparison



Interpretation

The table above reports Double Machine Learning estimates of the relationship between employment outcomes and three groups of predictors:

1. Health and Social Factors
2. Financial Responsibility
3. Lagged Wellbeing

For each treatment variable, the learner with the lowest nuisance-model MSE was selected as the preferred specification.

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